Greatest common factor

1 Find the greatest common factor between each set of terms:

a 3 and $15x$

b $8a$ and $9a$

c $33x$ and $6x^2$

d $12b$ and $15b^2x$

2 Complete the following statements:

a $4r + 12 = \square\,(r + 3)$

b $5v - 30 = \square\,(v - 6)$

c $18c - 24 = \square\,(3c - 4)$

d $2t^2 + 2t = 2t\,(\square + \square)$

3 Factor the following expressions:

a $6v + 30$

b $45t - 40$

c $-2s - 10$

d $-12s + 10$

e $y^2 + 4y$

f $k^2 - 5k$

g $2u^2 - 8u$

h $4t + 2t^2$

i $42x - x^2$

j $7x + 35$

k $36 - 6x$

l $-2x + 10$

m $-100 - 10x$

n $-18y^2 + 6$

o $9\,(3x + 4) + 4\,(3x + 4)$

p $8t\,(t - 3) + 9\,(t - 3)$

Grouping with 4 terms

4 Factor the following expressions:

a $5\,(a + b) + v\,(a + b)$

b $x\,(y - z) - w\,(y - z)$

c $8x\,(2y + 3w) - z\,(2y + 3w)$

d $8z\left(5x^2 + 4y\right) - \left(5x^2 + 4y\right)$

e $8y\,(y - 4) + 10\,(4 - y)$

f $(4c - d)\,(c + 9d) - 5\,(d - 4c)$

5 Determine whether or not the following expressions are in factored form:

a $-5\,(2y + z) - b\,(2y + z)$

b $(a + 7)\,(a + 2)$

c $(x - 5)\,(x + y)$

d $3x\,(a + 2b) + 2\,(a + 2b)$

6 Factor the following expressions:

a $x^2+5x+8x+40$

b $x^2-5x+10x-50$

7 Factor the following expressions:

a $xy-9x+2y-18$

b $56-8y-7x+xy$

c $12xy+wx+12yz+wz$

d $2x+xz-40y-20yz$

e $50+5y+10x+xy$

f $12xy+wx+12yz+wz$

8 By rewriting $4x^2+17x+4$ as an expression having four terms and factoring in pairs, factor the expression completely.

Trinomials

9 Complete the expressions that will make the following equations true.

a $(m+8)(\square)=m^2+18m+80$

b $y^2-8y-65=(y-13)(\square)$

10 Factor the following expressions:

a x^2+6x+5

b $x^2+14x+40$

c x^2-5x+4

d $x^2-13x+30$

e $x^2-15x+56$

f $x^2+x-132$

g $x^2+9x-90$

h x^2-2x-8

i $x^2-3x-70$

j $6+5x+x^2$

k $44-15x+x^2$

l $-12-13x-x^2$

m $-9+10x-x^2$

11 Factor the following expressions:

a $4x^2+56x+192$

b $-5x^2+10x+40$

c $3x^2-21x-54$

d $2x^3+16x^2+30x$

12 Complete the following equations to show the correct factors of the given expressions.

a $3x^2+10x-8=(3x-2)(x+\square)$

b $12x^2+25x+7=(4x+7)(3x+\square)$

c $15x^2-13x-6=(5x-\square)(3x+\square)$

13 Factor the following expressions:

a $7x^2 - 75x + 50$

b $6x^2 + x - 7$

c $9x^2 - 19x + 10$

d $7x^2 - 34x - 48$

e $6x^2 + 13x + 6$

f $-12x^2 - 7x + 12$

g $90 - 89x + 8x^2$

h $8 - 14p - 49p^2$

14 Factor the following expressions:

a $b^2 + 14b + 49$

b $y^2 - 10y + 25$

c $x^2 + 10x + 25$

d $x^2 + 14x + 49$

e $x^2 - 12x + 36$

f $x^2 - 16x + 64$

g $81x^2 - 72x + 16$

h $81x^2 + 36x + 4$

i $49x^2 - 28x + 4$

j $81 + 18x + x^2$

k $36 - 12x + x^2$

l $25 + 60r + 36r^2$

m $7d^2 + 112d + 448$

n $4x^2 + 40x + 100$

o $3x^2 - 24xy + 48y^2$

p $-6r^2 - 96r - 384.$

q $-3x^2 + 12x - 12$

r $b^2 - 7b + \dfrac{49}{4}$

s $x^2 + 3x + \dfrac{9}{4}$

t $x^2 - 3x + \dfrac{9}{4}$

Difference of squares

15 Write an expression that makes each of the following statements true.

a $p^2 - 64 = (p - 8)(\square)$

b $m^2 - 36 = (m - 6)\,(\square)$

16 Find the other factor of $x^2 - 49$ if one of its factors is $x + 7$.

17 Factor $x^4 - 100$ as a product of two binomials.

18 Consider the expression $81x^2 - 225$
Write the factored form of an equivalent expression, where a and b are integers.

19 Factor the following expressions:

- **a** $x^2 - 25$
- **b** $n^2 - 36$
- **c** $v^2 - 49$
- **d** $x^2 - \frac{4}{49}$
- **e** $81 - x^2$
- **f** $4 - u^2$
- **g** $4 - 49y^2$
- **h** $64x^2 - 121$
- **i** $121m^2 - 64$
- **j** $81x^2 - 100y^2$
- **k** $121x^2 - 49y^2$
- **l** $16x^2 - 25$
- **m** $6x^2 - 294$
- **n** $3t^2 - 12$
- **o** $5x^2 - 320$
- **p** $4n^2 - 16$